



CS335 Project Video Site

Team 20: Anjitha Joby, Kieran Hurley, Jannet Chen



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Credits

Anjitha Joby

- Wrote the user stories 8, 9, 10
- Placed sequence diagrams about Video Editing, Content Recommendations & Cloud Storage
- Wrote class diagram slide
- Created Test Cases 8, 9, 10
- Created first Test Plan slide
- Created EP + BVA Table slide

Kieran Hurley

- Created Title slide
- Wrote User Stories 1,2,3,4
- Placed Sequence Diagrams for Commenting, Searching, Following and Channel Analytics
- Wrote Architecture diagram slide
- Created Test Cases 1,2,3,4

Jannet Chen

- Wrote User Stories 5,6,7
- Placed sequence diagram Thumbnail Generation, Voting, Video Sharing
- Wrote Use Case diagram slide
- Created Test Case 5,6,7
- Created second Test Plan slide



User Stories

- 1.As a registered user I want to leave a comment underneath the video I'm currently watching so that I can share my opinion for other users and the creator to see. Clicking the comment button should allow the user to view other comments left by other users, sort comments by 'popular' and 'recent' and vote on other user comments and reply.
- 2.As a registered user I want to be able to search for specific videos and users so that I can find content that caters to my current needs. Search predictions based on user likes will appear as they type to help speed up the search process, with results sorted by popularity and positive voting..
- 3.As a registered user, I want to be able to follow certain creators so that I can view their uploads and be updated when they make announcements or posts. If the user decides to follow a creator, they will be prompted to enable notifications and once followed, the creator will be recommended more frequently in their feed.
- 4.As a registered user, I want to be able to view my videos analytics to improve on my content and expand my userbase. Each video will be rated positive or negative to quickly brief the creator on performance, with detailed analytics including views, comments, likes and like-to-dislike ratio accessible via the three-dot button.
- 5.As a registered user, I want to generate and edit video thumbnails to attract viewers and maintain creative control. This allows users to create and modify video thumbnails, helping them make their content more attractive and maintain creative control.

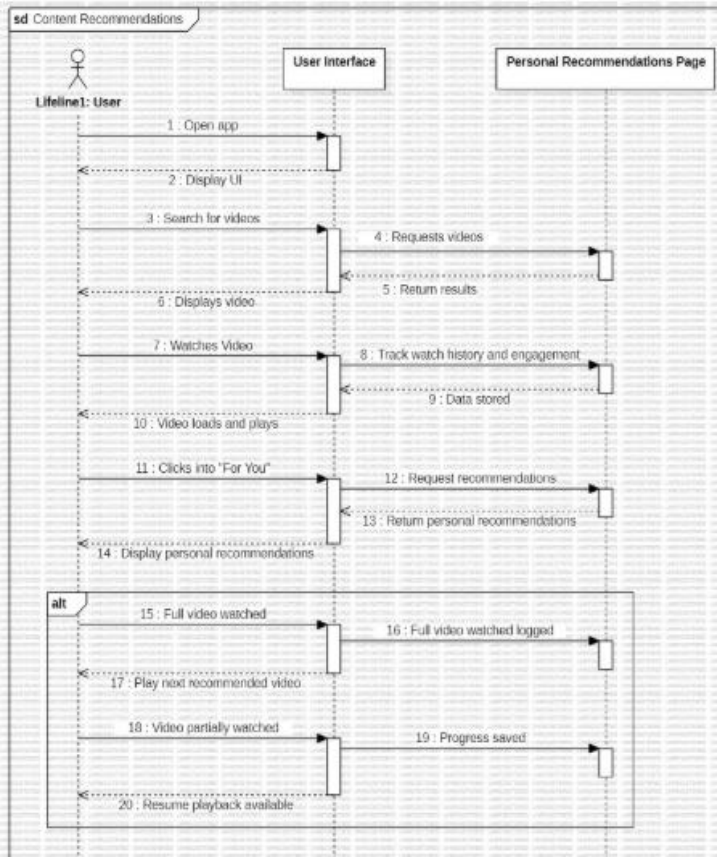


User Stories

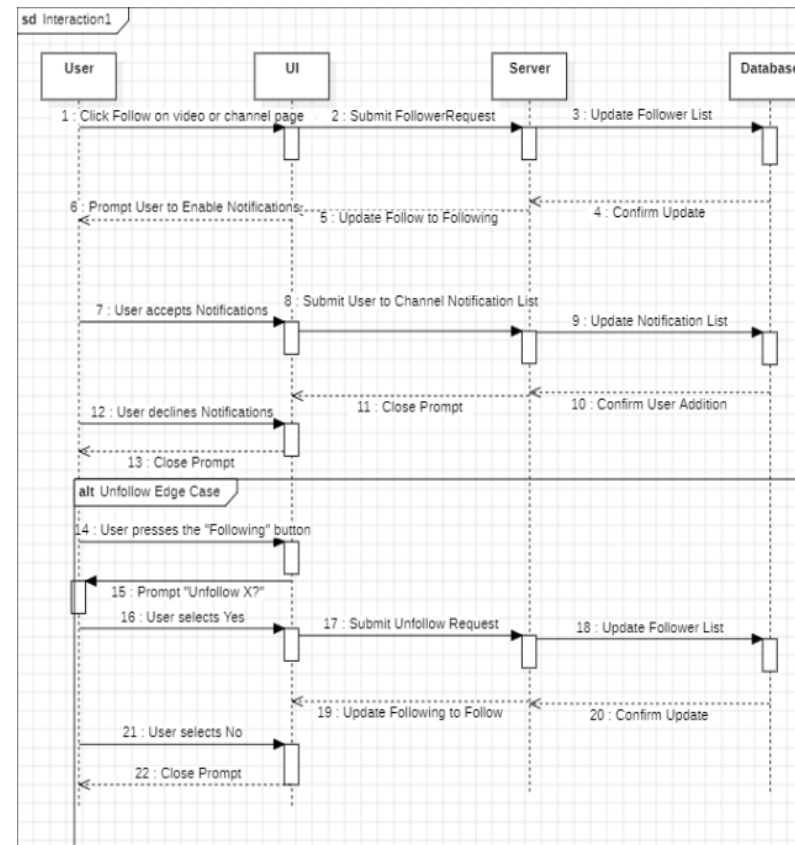
6. As a registered user, I want to vote on videos and polls to express my preferences and see real-time engagement. This enables users to vote on videos and polls, allowing them to express opinions and see real-time engagement from others.
7. As a registered user, I want to share videos across platforms with proper privacy and tracking to maximize reach and interaction. This feature lets users share videos across different platforms while controlling privacy and tracking interactions to increase reach.
8. As a registered user, I want to be able to create a video where I can import video clips and edit them using different editing tools. This will give me an opportunity to create my own videos by combining and editing clips, allowing me to export a final piece.
9. As a registered user, I want to watch videos based on my watch history, so I can watch content that matches my personal interests. This allows me to watch personalized videos without having to search for them, improving my experience overall.
10. As a registered user, I want to upload videos and drafts to the cloud, so that I can easily access them to prevent losing any work. This will allow me to continue editing, with the opportunity to access my work on different devices.

Sequence Diagrams

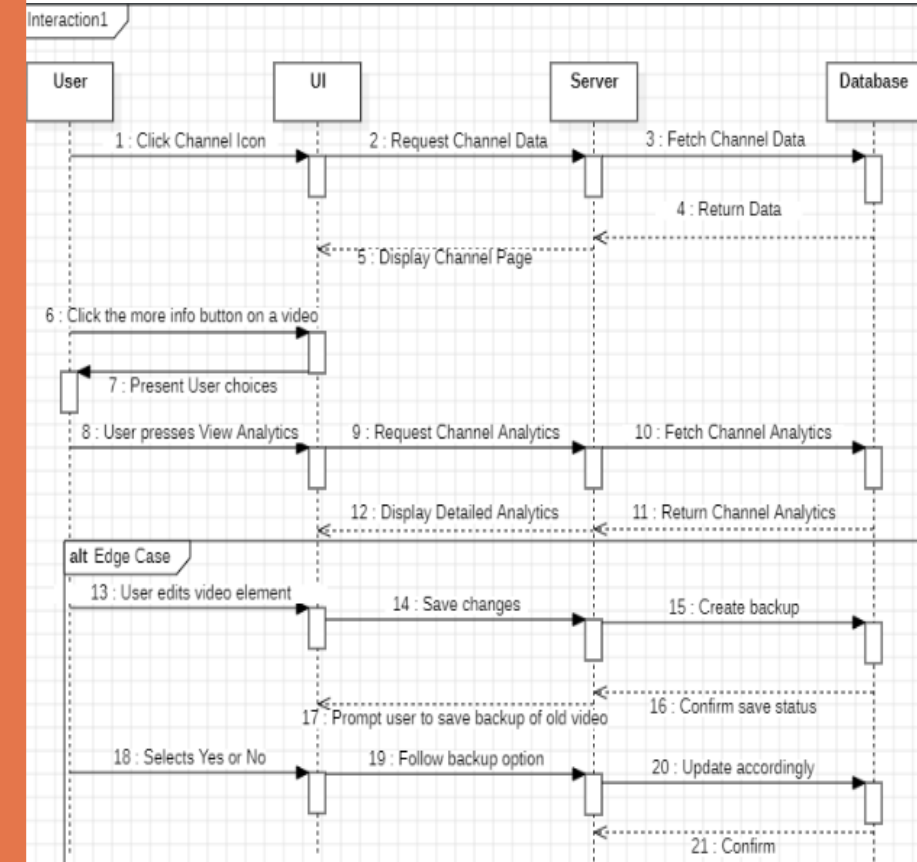
Content Recommendations Diagram:



Follow Diagram:

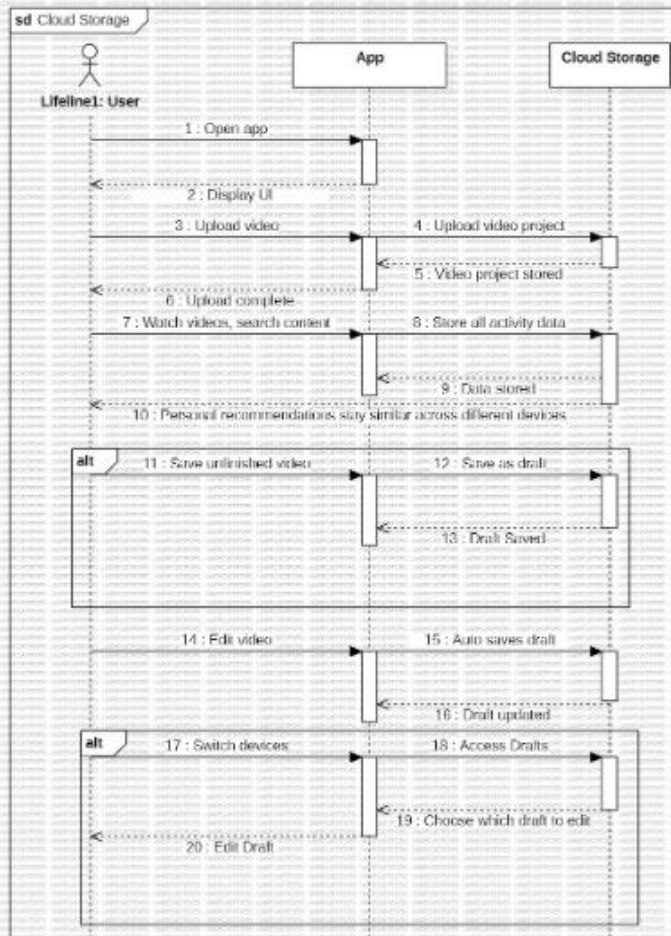


Video Analytics Diagram

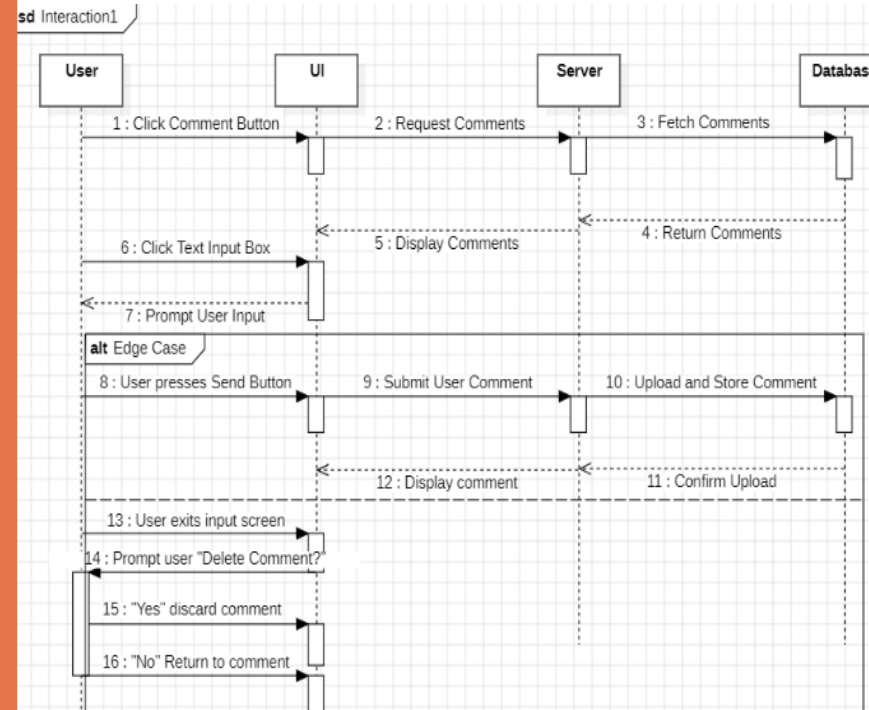


Sequence Diagrams

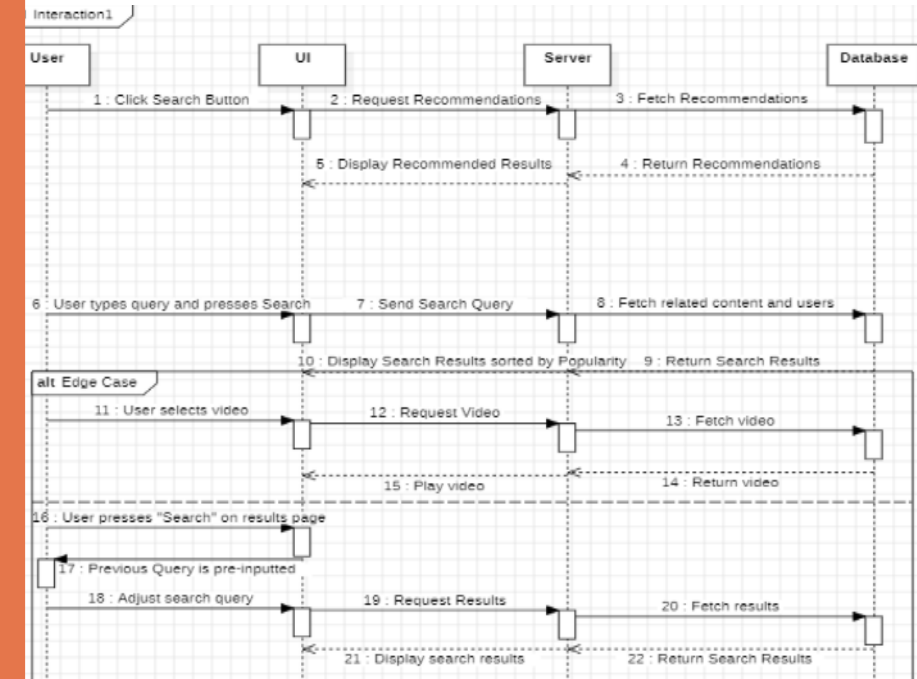
Cloud Storage Diagram:



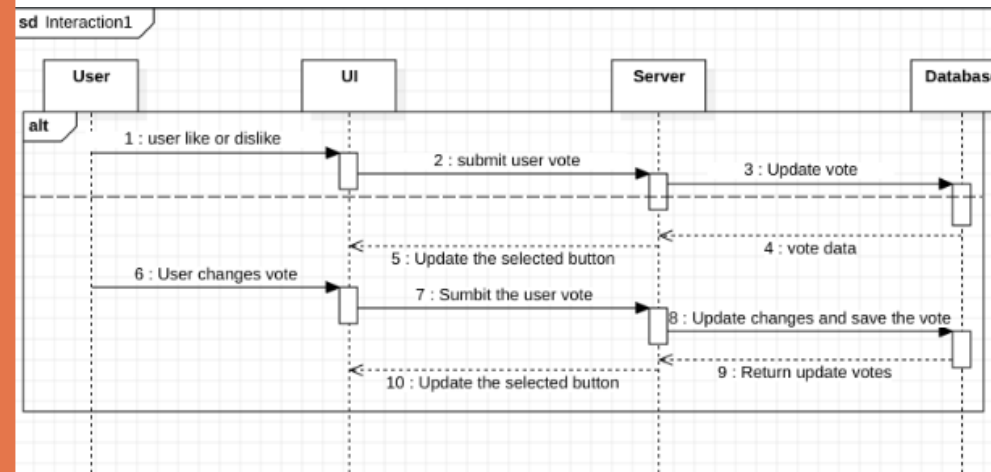
Comment Diagram:



Search Diagram:

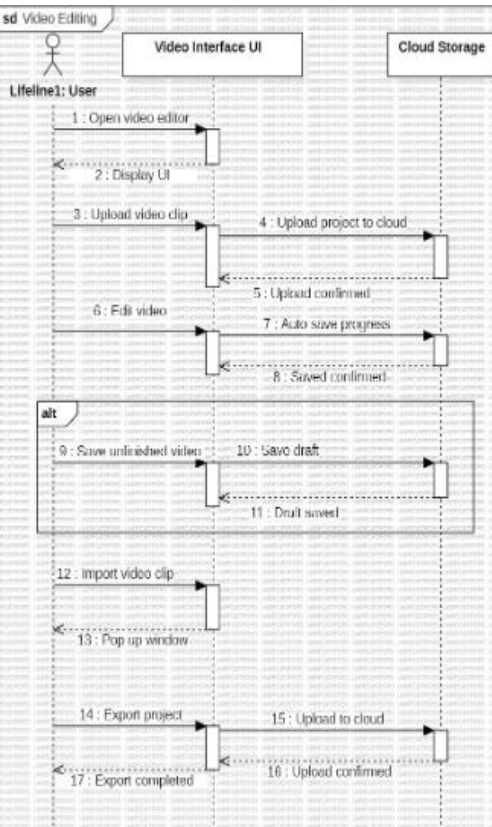


Voting Diagram:

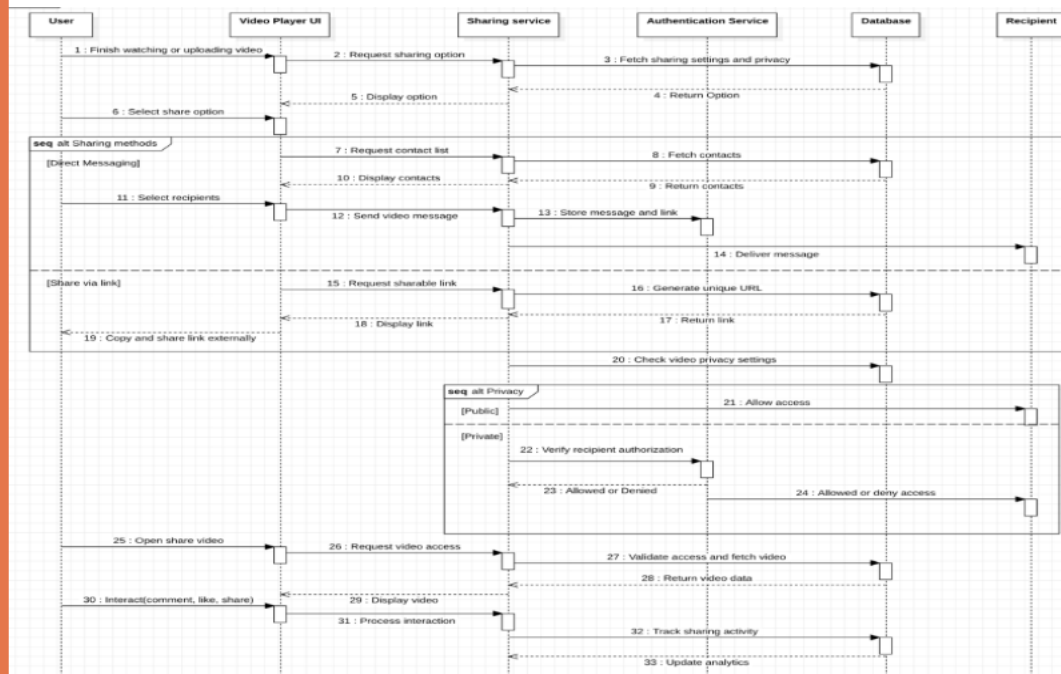


Sequence Diagrams

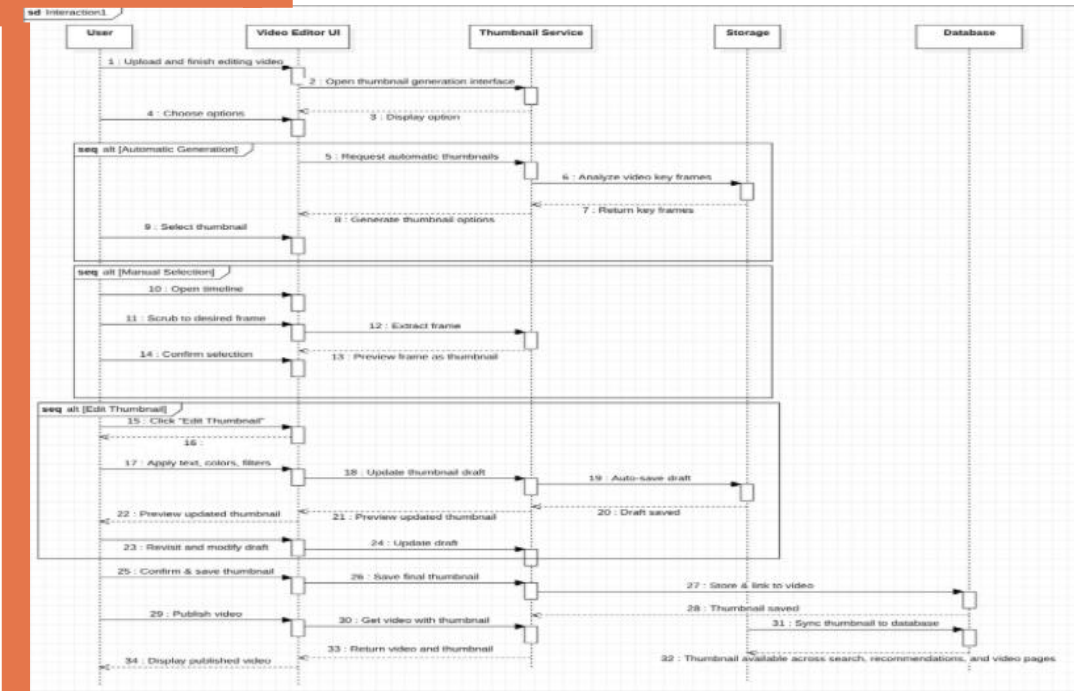
Video Editing Diagram:



Video Sharing Diagram:

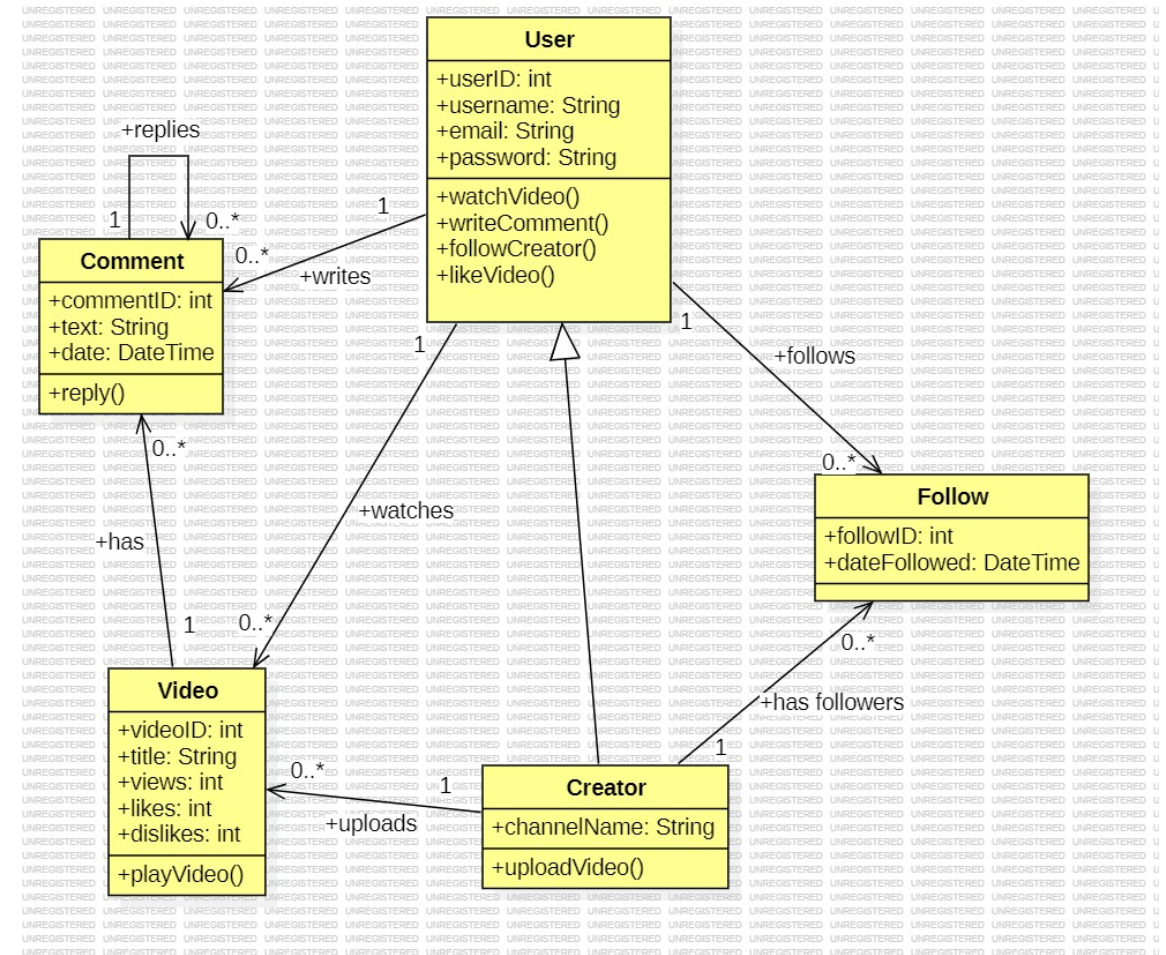


Thumbnail Generation Diagram:



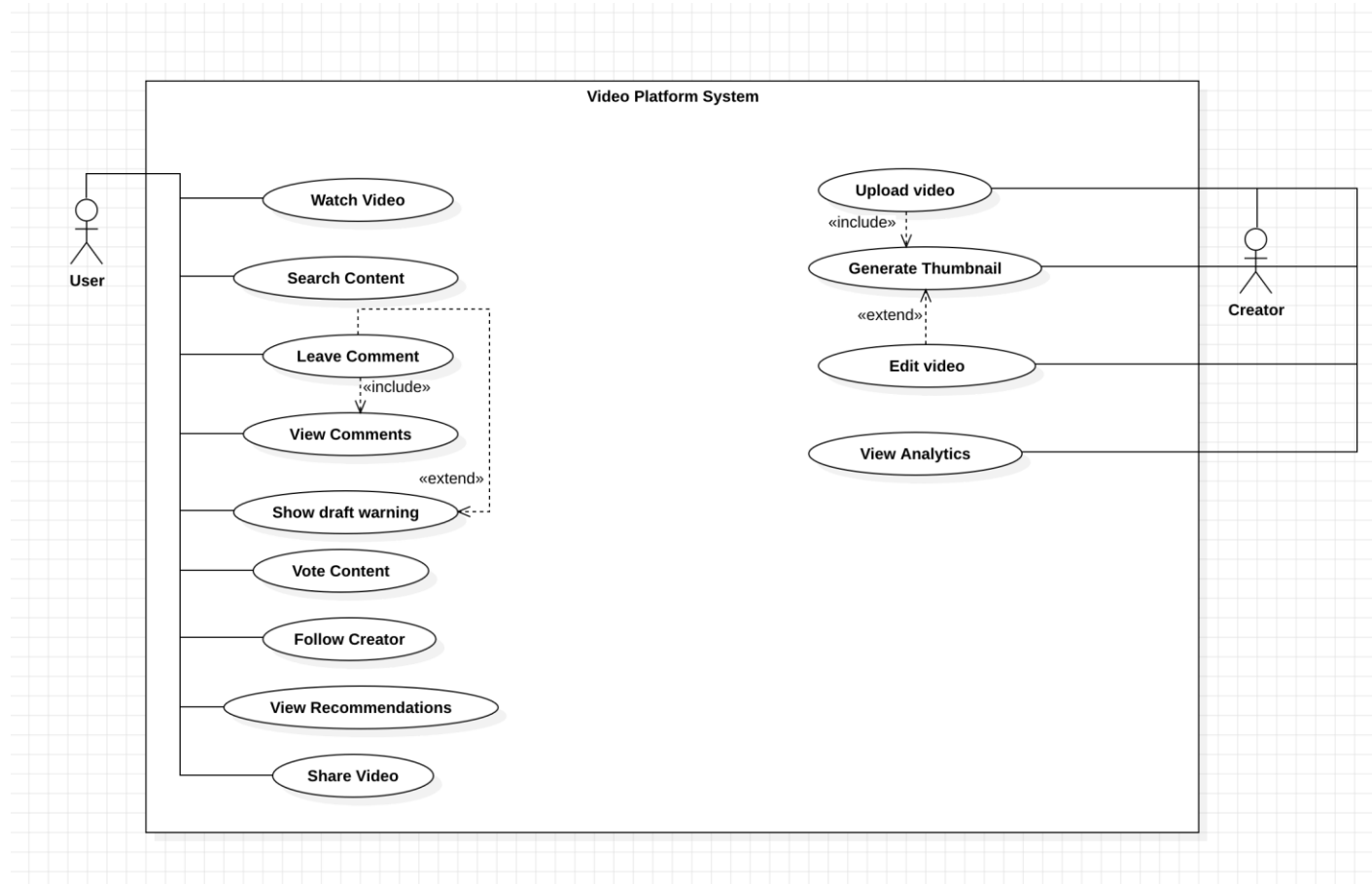
Class Diagram

- Outlines the database for a content creation app.
- Shows users who watches and interacts with videos
- Shows content creator who uploads the videos.
- Users can follow Creators, watch videos and post comments on videos.
- Video is uploaded by the Creator and the views, likes, dislikes are tracked.



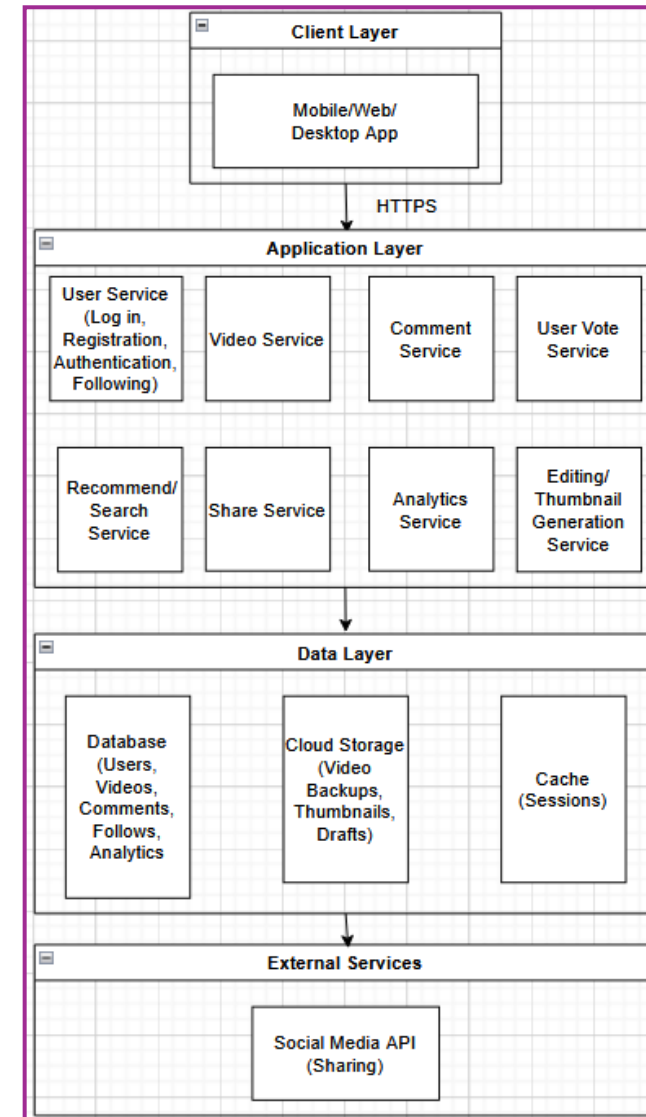
Use Case Diagram

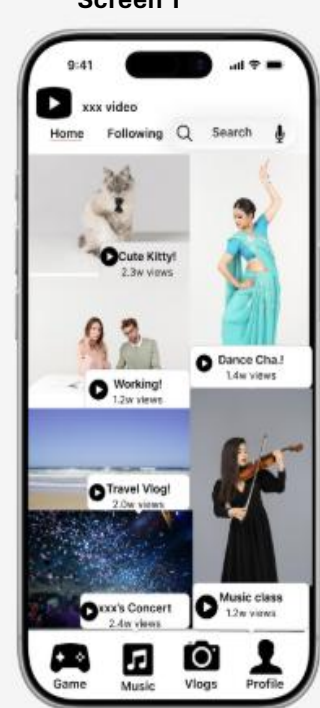
- Users can **watch videos**, **search content**, and **view recommendations**.
- Users can **leave comments**, **view comments**, and receive **draft warnings**.
- Users can **vote on content**, **follow creators**, and **share videos**.
- Creators can **upload videos** and **edit videos**.
- Creators can **generate thumbnails** (included in upload process).
- Creators can **view analytics** to track performance.



Architecture Diagram

- Represents a high-level architecture for the video site.
- Client Layer: this is what the user interacts with directly to use the app. (E.g. Mobile App)
- Application Layer: This contains the necessary services required for the app to function
- Data Layer: This stores data necessary for the application layer to function, allowing data to be received, transferred, altered and deleted.
- External Services: Allows use for off-platform sharing

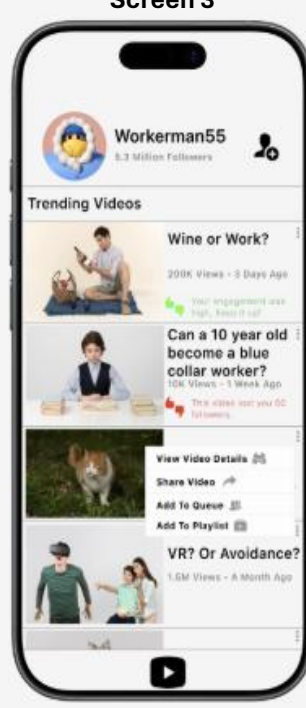




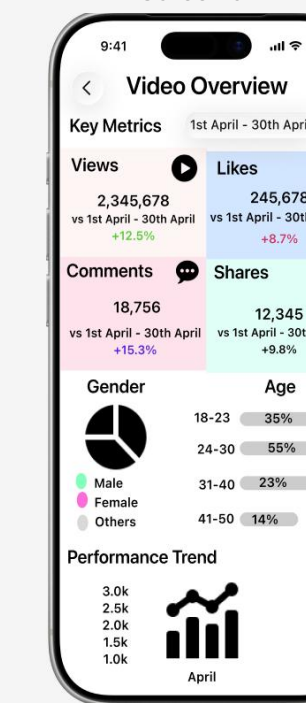
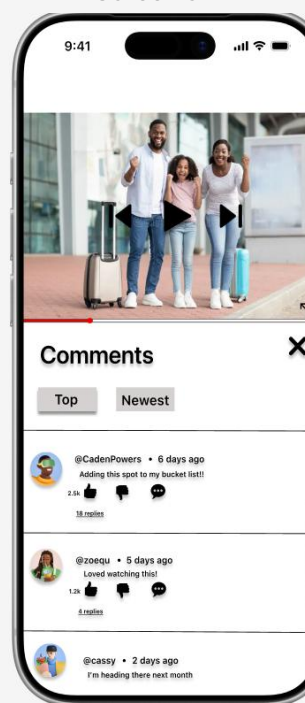
Screen 4



Screen 5



Screen 6



UI Mock-Up

Screen 1: Home Screen

- Top navigation: Home / Following / Search
- Bottom menu: Game, Music, Vlogs, Profile
- Supports content discovery and quick browsing

Screen 2: Video Player

- Top of screen displays video window
- Below Video player, the title, views and date the video is posted is directly below.
- Controls below video: Open Channel Page, Follow, Like, Dislike, Comment
- Below are recommend videos to watch
- Bottom menu: Home button

Screen 3: Channel Page

- Top of screen displays channel name and profile picture.
- Top controls: Follow
- Trending videos from user are listed with details and analytics displayed.
- Video controls: ⋮ is a more button
- More button allows user to access more options.

Screen 4: Video Editor

- Preview window shows current frame of the window
- Top controls: Close, Search, Export
- Playback controls such as play/pause and undo/redo
- Timeline shows video clip where users can add text/audio/effects
- "+" button to add new clips
- Bottom toolbar contains editing tools

Screen 5: Comment Section

- Displays top user's comments
- Can scroll down to view more comments
- Users interact by liking/disliking and adding a response to other user's comments
- Comment posted by users displays their profile picture and username

Screen 6: Video Overview

- Key metrics: Views, Likes, Comments, Shares
- Time-based comparison (growth/decline trends)
- Audience insights: Age and gender distribution
- Performance trend chart

Test Plan

Introduction:

- This test plan will cover the testing of a video sharing and editing application, which is designed for registered users and creators. The testing will focus on the 10 user stories. This will make sure all requirements are met and all errors are handled.

Scope:

In Scope:

- All user stories will be tested for registered users and creators.
- Commenting on videos
- Searching for videos
- Following creators
- Viewing video analytics
- Thumbnail Editing
- Voting on videos (like/dislike)
- Sharing videos
- Video Editing
- Personalised recommended videos based on watch history
- Uploading videos to the cloud

Test Plan

Out of Scope:

- Security features
- Creator monetisation features
- Subscription plans
- Copyright
- Payment and billing

Approach:

- The testing approach for this project combines manual, functional, unit, and system testing to ensure all user stories and features work correctly. Both positive and negative test cases are used to validate expected behavior and error handling across features such as commenting, searching, following, analytics, thumbnail generation, voting, sharing, video editing, recommendations, and cloud uploads.
- Equivalence Partitioning (EP) and Boundary Value Analysis (BVA) are applied to input-based features, such as testing comment limits with values like 0, 1, 10,000, and 10,001 characters, to ensure correct handling of valid and invalid inputs. Unit testing focuses on smaller components like input validation and file checks, while system testing verifies complete user workflows. All results are recorded as Pass or Fail, and any issues are retested after fixes to ensure the system meets all requirements and acceptance criteria.

Test Cases

Test case 1:

Field	Description
Test Case ID	TC-COMMENT-01
Title	As a User I want to leave a comment
Preconditions	User is already logged in and watching video.
Steps	1. Press Comment button 2. Tap Text Input Box 3. Type and send comment. (With stable internet connection)
Expected Result	Comment is displayed on User's screen and visible for other users
Pass/Fail	

User Story:	"As a User I want to leave a comment"
Acceptance Criteria:	Comment Displayed

TC	STEPS	EXPECTED RESULT
TC-COMMENT-01	Type and upload comment (With internet)	Comment is displayed on User's screen and visible for other users
TC-COMMENT-02	Type comment and attempt to back out.	User is prompted to confirm they'll discard their drafted comment.
TC-COMMENT-03	Type comment over 10,000 characters	User is over character <u>limit</u> , error is thrown and told to reduce comment size.

Test case 2:

Field	Description
Test Case ID	TC-SEARCH-01
Title	As a User I want to search for specific videos.
Preconditions	User is already logged in.
Steps	1. Press Search button 2. Tap Text Input Box 3. Type Search Query 4. Confirm search (Query has matching results)
Expected Result	The user is displayed the appropriate content for the query.
Pass/Fail	

User Story:	"As a User, I want to search for a specific video"
Acceptance Criteria:	Search shows matching results "No Results" if nothing matches

TC	STEPS	EXPECTED RESULT
TC-SEARCH-01	Search for a video (Query has valid results)	The user is displayed the appropriate content for the query.
TC-SEARCH-02	Search for a video (Query has no valid results)	The user is told their query yields no valid results.
TC-SEARCH-03	Search for a video and edit query	User's previous query should still be present.

Test case 3:

Field	Description
Test Case ID	TC-FOLLOW-01
Title	As a User I want to follow my favourite creators.
Preconditions	User is already logged in, User is on either a video or channel page with a follow button present
Steps	1. Press the follow button 2. Select whether to enable notifications
Expected Result	The follow button updates from "Follow" to "Following"
Pass/Fail	

User Story:	"As a User, I want to follow my favourite creators"
Acceptance Criteria:	Button shows "Following" Button shows "Follow" if user chooses to unfollow

TC	STEPS	EXPECTED RESULT
TC-FOLLOW-01	The user presses the "Follow" button. (They are not currently following)	After deciding notification preference, "Follow" should update to "Following"
TC-FOLLOW-02	The user presses the "Follow" button. (They are currently following)	After deciding whether to confirm unfollowing the creator, "Following" should update to "Follow"
TC-FOLLOW-03	The user follows a deactivated creator.	Error message shown, "Follow" stays unchanged

Test case 4:

Field	Description
Test Case ID	TC-ANALYTICS-01
Title	As a User I want to view my video analytics to make improvements.
Preconditions	User is already logged in, User is on their channel page, User has uploaded videos
Steps	1. Press the 3 dotted button on any video 2. Press view video details 3. Open video overview
Expected Result	An array of video results <u>are</u> presented to the user.
Pass/Fail	

User Story:	"As a User I want to view my video analytics to make improvements."
Acceptance Criteria:	Any form of analytics being displayed Insufficient Data

TC	STEPS	EXPECTED RESULT
TC-ANALYTICS-01	The user views their video and opens their video overview (Video has data)	User should be provided the appropriate analytics
TC-ANALYTICS-02	The user views their video and opens their video overview (Video has insufficient data)	The user should be told their video has insufficient data for analytics.
TC-ANALYTICS-03	The user views their video and opens their video overview (Video has been taken down)	The user should be told that his video no longer exists.

Test Cases

Test case 5:

Field	Description
Test Case ID	TC-Thumbnail Generation and Editing-01
Title	Generate automatic video thumbnail
Preconditions	User is logged in as a creator and has upload a video
Steps	1.Complete video upload 2. Open thumbnail generation interface 3. Select” Auto-Generate-Thumbnail”
Expected Result	System generates multiple thumbnail options from key frames.
Pass/Fail	

User Story:	“As a register creator, I want to generate and edit video thumbnails”
Acceptance Criteria:	User can generate thumbnails automatically.

TC	STEPS	EXPECTED RESULT
TC- Thumbnail Generation and Editing-02	1.View generates thumbnails 2.Select one thumbnails 3.Confirm selection	Selected thumbnail is saved and applied to video.
TC- Thumbnail Generation and Editing-03	1.Select a thumbnail 2.Edit text/layout/colours 3.Save change	Selected thumbnail is saved and applied to video
TC- Thumbnail Generation and Editing-04(neg)	1. Edit thumbnail 2. Exit without saving	System prompts user to save or discard changes

Test case 6:

Field	Description
Test Case ID	TC-Video Sharing-01
Title	Share video via link
Preconditions	User is logged in and watching a video.
Steps	1.Click” Share” button 2. Select “Copy link” 3. Open the link in browser
Expected Result	Video opens successfully with correct permissions.
Pass/Fail	

User Story:	“As a register creator, I want to share videos across platforms”
Acceptance Criteria:	User can share via link or direct message.

TC	STEPS	EXPECTED RESULT
TC-Video Sharing-01	1. Click share 2. Select user/contact 3. Send	Recipient receives video link.
TC- Video Sharing-02	1. Share private video link 2. Open with unauthorized user	Access is denied.
TC- Video Sharing-03	1. Click share 2. Disconnect internet 3. Attempt to share	Error messages are displayed.

Test case 7:

Field	Description
Test Case ID	TC-Voting-01
Title	Vote on a video
Preconditions	User is logged in and watch a video
Steps	Click “like” or “dislike”
Expected Result	Vote is recorded and updated in real time
Pass/Fail	

User Story:	“As a register creator, I want to vote on videos and polls”
Acceptance Criteria:	User can like or dislike the video.

TC	STEPS	EXPECTED RESULT
TC-Voting-01	1. Select poll options 2. submit vote.	Poll results are displayed with updated percentages.
TC-Voting-02	1.Vote on video 2. Attempt to vote again	System prevents multiple voting.
TC- Voting-03	1. Vote on a poll 2. Change selection (if allowed)	Vote updates successfully.

Test Cases

Test case 8:

Field	Description
Test Case ID	TC-VIDEOEDITOR-01
Title	User wants to import and edit the video
Preconditions	User logged into account, video editor open, video clips imported
Steps	1. Import video clips 3. Apply editing tools 4. Preview the video
Expected Result	Videos successfully imported, editing tools applied, video preview shown
Pass/Fail	

User Story:	As a user, I want to import and edit video clips using different editing tools.
Acceptance Criteria:	Successfully import videos into the video editor

TC	STEPS	EXPECTED RESULT
TC-01	User imports and edits the video	Video is imported, edits are reflected in the preview
TC-02	User edits without importing a video	Message display "Import video to start"
TC-03	User imports unsupported file type	Video not imported, Message displays "Unsupported file format"

Test case 9:

Field	Description
Test Case ID	TC-CONTENTRECOMMENDATIONS-01
Title	User wants to see more personal recommendations based on their watch history
Preconditions	User is logged in, user has previously watched videos, watch history recorded
Steps	1.Open app 2. Click into home page 3. Click into a video
Expected Result	User recommended videos displayed based on watch history
Pass/Fail	

User Story:	As a user, I want to watch recommended videos that <u>matches</u> my personal interests.
Acceptance Criteria:	User watch history should be recorded to display user recommendations

TC	STEPS	EXPECTED RESULT
TC-01	Select into home page (watch history exists)	User is shown <u>personalised</u> recommended videos
TC-02	Select into home page (no watch history)	<u>User is</u> shown trending videos.

Test case 10:

Field	Description
Test Case ID	TC-UPLOAD-01
Title	User wants to upload videos to the cloud
Preconditions	User account exists, system is running, video editor open,
Steps	1.Select "Export" 2. Select "Upload to cloud"
Expected Result	Video successfully uploaded and backed up in cloud. User receives confirmation message.
Pass/Fail	

User Story:	As a user, I want to upload my videos to the cloud.
Acceptance Criteria:	Successfully upload videos to the cloud.

TC	STEPS	EXPECTED RESULT
TC-01	Select "Export" and then select "Upload to cloud" (With Internet and Storage)	Video successfully backed up in cloud and user receives confirmation message
TC-02	Select "Export" and then select "Upload to cloud" (With No Internet)	Video backup failed with error message "No internet connection"
TC-03	Select "Export" and then select "Upload to cloud"" (With No Storage)	Video backup failed with error message "No storage available"

EP + BVA Table

Test No.	Test Cases Covered	Inputs	Expected Outputs
1	TC-COMMENT-01	100	True
2	TC-COMMENT-03	11,000	False
3	TC-COMMENT-04*	0	False
4	TC-COMMENT-05*	1	True
5	TC-COMMENT-06*	2	True
6	TC-COMMENT-07*	9,999	True
7	TC-COMMENT-08*	10,000	True
8	TC-COMMENT-09*	10,001	False

Scenario	Test Step	Expected Result	Actual Outcome	Level
Verify that the input field accepts comment within limit	Enter 100 characters and upload	Application should be able to accept all 100 characters comment	Application accepts all 100 characters	System
Verify that the input field rejects comment over the limit	Enter 11,000 characters and upload	Application should not accept all 11,000 characters comment and show error	Application rejects comment, shows error to reduce comment size	System
Verify that the input field rejects empty field	Enter 0 characters and upload	Application should reject empty comment	Application rejects empty comment, shows error	Unit
Verify that the input field accepts comment of minimum of 1 character	Enter 1 character and upload	Application should be able to accept all 1 character comment	Application accepts all 1 character	Unit
Verify that the input field accepts comment just above minimum of 2 characters	Enter 2 characters and upload	Application should be able to accept all 2 characters comment	Application accepts all 2 characters	Unit
Verify that the input field accepts comment just below maximum of 9,999 characters	Enter 9,999 characters and upload	Application should be able to accept all 9,999 characters comment	Application accepts all 9,999 characters	Unit
Verify that the input field accepts comment with maximum of 10,000 characters	Enter 10,000 characters and upload	Application should be able to accept all 10,000 characters comment	Application accepts all 10,000 characters	Unit
Verify that the input field rejects comment over maximum of 10,001 characters	Enter 10,001 characters and upload	Application should not accept all 10,001 characters comment and show error	Application rejects comment, shows error to reduce comment size	Unit